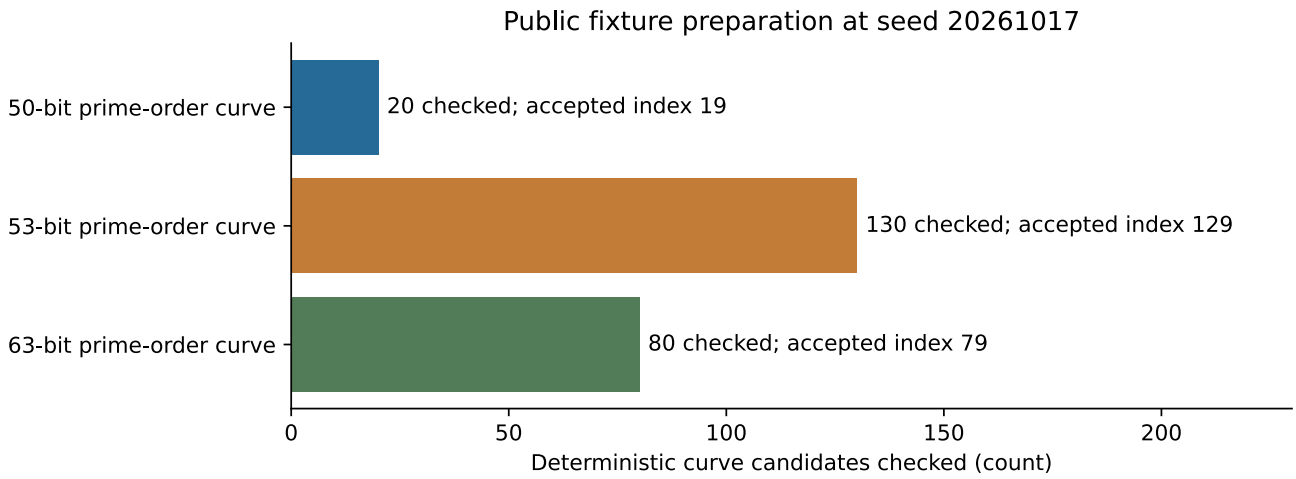
Certified public prime-field fixtures for ICPERF

Engineering preparation, 2026-10-17 UTC. Three public prime-order curves were selected before any factor base or target. These are curve fixtures, not index-calculus trials or speed measurements. The JSON records and source are linked from the companion Markdown report.



Source: research/icperf_prime_fixtures_20261017/n{50,53,63}_seed20261017.json. This is selection effort, not IC or rho cost.

The chart counts deterministic candidate curves checked at seed 20261017. At 53 bits a 128-candidate search ended without an accepted curve; the same sequence extended to candidate index 129. That bounded result says nothing about other 53-bit curves.

Exact accepted identities

Bits	Checked	Field prime p	Certified order N
50	20	1125899798684033	1125899778887903
53	130	9007198804718503	9007198733080069
63	80	9223372021248595841	9223372018953067627

Each record includes the exact a, b, affine witness, candidate prefix, source SHA-256 and runtime provenance. Source commit: `d5a7a863f34ec321623cd56275243b0a7a851d1a`.

Independent certificate

The plain-Python verifier checks p and N prime, a nonsingular curve, an affine nonidentity witness P on it, N P = O, the exact Hasse inequality `(N-(p+1))² <= 4p`, and `N² > 16p`. Since N is prime, P has order N. Thus N divides the curve order; the Hasse interval contains N and is narrower than N, so the curve order equals N. This check does not call Sage cardinality. Sage proof-mode primality also returned true for p and N in all three records.

What remains

Factor bases, point decomposition for m=2..8, direct S8 capacity, relation yield, useful rank, linear algebra, target recovery, and same-point rho have not been measured on these curves. The prime implementation's compact scan currently needs p below 2^32; a wide exact path or a charged fallback is the next implementation gate. Base selection must stay target-blind, and its construction and rejected candidates belong in cold-start cost.

The canonical ECDLP frontier graph was checked. It is unchanged because this preparation adds no reviewed attack-cost finding.